magnetic fields. [Ch. 9]

Alexandrov and Pimenov initiate an ill-fated attempt to introduce topological tools into mathematical studies of curved spacetime. [Ch. 13]

1951 Teller and Vlam in the U.S. invent the idea for a "real" superbomb, one that can be arbitrarily powerful; Wheeler puts together a team to design a bomb based on the idea and simulate its explosion on computers. [Ch. 6]

Graham Smith provides Baade with a 1-arc-minute error box for the cosmic radio source Cyg A, and Baade discovers with an optical telescope that Cyg A is a distant galaxy—a "radio galaxy." [Ch. 9]

1952 The U.S. explodes its first superbomb device, one too massive to be delivered by an airplane or rocket, but using the Teller–Vlam invention and based on the Wheeler team's design work. [Ch. 6]

1953 Wheeler launches into research on general relativity. [Ch. 6]

Jennison and Das Gupta discover that the radio waves from galaxies are produced by two giant lobes on opposite sides of the galaxy. [Ch. 9]

Stalin dies. [Ch. 6]

The U.S.S.R. explodes its first hydrogen bomb, based on the Ginzburg and Sakharov ideas. It is claimed by U.S. scientists not to be a "real" superbomb because the design does not permit the bomb to be arbitrarily powerful. [Ch.6]

1954 Sakharov and Zel'dovich invent the Teller–Ulam idea for a "real" superbomb. [Ch. 6]

The U.S. explodes its first real superbomb, based on the Teller–Ulam/Sakharov–Zel'dovich idea. [Ch. 6]

Teller testifies against Oppenheimer, and Oppenheimer's security clearance is revoked. [Ch. 6]